

Problem

An *averaging amplifier* is a summer that provides an output equal to the average of the inputs. By using proper input and feedback resistor values, one can get

$$-v_{\text{out}} = \frac{1}{4}(v_1 + v_2 + v_3 + v_4)$$

Using a feedback resistor of 10 k Ω , design an averaging amplifier with four inputs.

Step-by-step solution

Step 1 of 4

The desired expression of the averaging amplifier is as follows:

$$-v_{\text{out}} = \frac{1}{4}(v_1 + v_2 + v_3 + v_4) \dots\dots (1)$$

Refer to equation (5.15) in the textbook for the expression for the output of a summing amplifier with three different input voltages, v_1 , v_2 , and v_3 .

Rewrite the expression for the output of a summing amplifier with four different input voltages, v_1 , v_2 , v_3 , and v_4 as follows:

$$v_{\text{out}} = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \frac{R_f}{R_3}v_3 + \frac{R_f}{R_4}v_4\right) \dots\dots (2)$$

Here, R_f is the feedback resistor, R_1 , R_2 , R_3 , and R_4 are the resistors in series with the input voltages v_1 , v_2 , v_3 , and v_4 respectively.

Step 2 of 4

Since value of feedback resistor R_f is $10\text{ k}\Omega$, substitute $10\text{ k}\Omega$ for R_f in equation (2).

$$v_{\text{out}} = -\left(\frac{10 \times 10^3}{R_1} v_1 + \frac{10 \times 10^3}{R_2} v_2 + \frac{10 \times 10^3}{R_3} v_3 + \frac{10 \times 10^3}{R_4} v_4\right)$$

$$= -10 \times 10^3 \left(\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} + \frac{v_4}{R_4}\right)$$

In order to have 4 in the denominator, the values of resistors R_1, R_2, R_3 , and R_4 must be equal, that is, R . thus, v_{out} becomes,

$$v_{\text{out}} = -10 \times 10^3 \left(\frac{v_1}{R} + \frac{v_2}{R} + \frac{v_3}{R} + \frac{v_4}{R}\right)$$

$$= -\frac{10 \times 10^3}{R} (v_1 + v_2 + v_3 + v_4) \dots\dots (3)$$

Compare equation (1) and equation (3).

$$\frac{10 \times 10^3}{R} = \frac{1}{4}$$

$$R = 40 \times 10^3 \Omega$$

$$= 40\text{ k}\Omega$$

Therefore,

$$R_1 = R_2$$

$$= R_3$$

$$= R_4$$

$$= 40\text{ k}\Omega$$

Step 3 of 4

Draw the amplifier using $R_f = 10\text{ k}\Omega$, four inputs v_1, v_2, v_3 , and v_4 , and resistors R_1, R_2, R_3 , and R_4 equal to $40\text{ k}\Omega$, as shown in Figure 1.

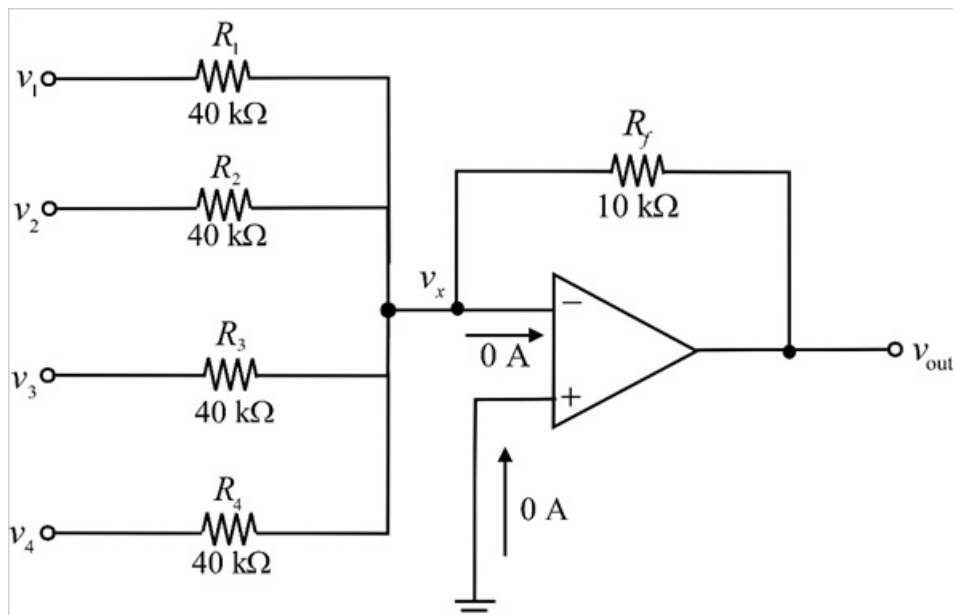


Figure 1

Step 4 of 4

Verify the circuit using the numerical calculations.

Apply Kirchhoff's current law at node v_x .

$$\frac{v_1 - v_x}{40 \times 10^3} + \frac{v_2 - v_x}{40 \times 10^3} + \frac{v_3 - v_x}{40 \times 10^3} + \frac{v_4 - v_x}{40 \times 10^3} = \frac{v_x - v_{out}}{10 \times 10^3}$$

$$\frac{1}{40 \times 10^3} (v_1 + v_2 + v_3 + v_4) + v_x \left(\frac{4}{40 \times 10^3} - \frac{1}{10 \times 10^3} \right) = \frac{-v_{out}}{10 \times 10^3}$$

Since $v_x = 0$ V because of virtual short circuit between inverting and non-inverting terminals, therefore,

$$\frac{1}{40 \times 10^3} (v_1 + v_2 + v_3 + v_4) + 0 \left(\frac{4}{40 \times 10^3} - \frac{1}{10 \times 10^3} \right) = \frac{-v_{out}}{10 \times 10^3}$$

$$\frac{1}{40 \times 10^3} (v_1 + v_2 + v_3 + v_4) = \frac{-v_{out}}{10 \times 10^3}$$

$$-v_{out} = \frac{10 \times 10^3}{40 \times 10^3} (v_1 + v_2 + v_3 + v_4)$$

$$-v_{out} = \frac{1}{4} (v_1 + v_2 + v_3 + v_4) \dots\dots (4)$$

The expression obtained in equation (4) matches with the desired expression in equation (1).

Hence, the averaging amplifier with four inputs have been designed.